DO JAPANESE PREFECTURES START NEW INDUSTRIES THAT USE THEIR WORKERS' SKILLS? A SKILL-NETWORK STUDY

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1. Problem and gap

Countries and regions tend to move into activities related to what they already do (Hidalgo et al., 2007; Neffke et al., 2011). Industries can be related because they locate together, trade with each other, or need the same skills. Neffke et al. (2017) measure the third from German job changers: moves cluster in a few industry pairs, forming a sparse, stable network that predicts where industries enter and grow better than location or trade links. Diodato et al. (2018) add, for the US, that shared skills now explain why services cluster, while supply chains still matter in manufacturing. Others use occupations: Buyukyazici et al. (2024) find Italian regions specialise in industries that fit local workers' skills, and Deegan et al. (2024) show related occupations help Norwegian regions move into complex jobs. Network structure matters too. Abbasiharofteh et al. (2023) find more unusual inventions where links connect inventor groups with different specialisations.

Most Japanese studies of regional capabilities use firms or patents, not workers. Chakraborty et al. (2020) link 47 prefectures to 91 sectors using over one million firms and find more complex prefectures are richer. Karashima and Inoue (2026, preprint) find prefectures grow faster after their patents become more sophisticated. Worker-based evidence is thin. Tomiura (2003) finds manufacturing employment grew faster where industries with similar occupation mixes clustered, but uses only three occupation groups. Kondo and Naganuma (2015) show Japanese workers tend to avoid industries needing different tasks and favour industries that trade with their old one, but build no network for prefectures.

Gap statement: To our knowledge, Japan has no detailed skill-relatedness network, from job changes or occupations. So we do not know whether Japanese prefectures move into industries that use their workers' existing skills. This matters because prefectures and training programmes choose which industries to support, and their plans fail if local workers lack the needed skills.

2. Research question and hypothesis

RQ: Are Japanese prefectures more likely to start new industries that need the same skills as the industries they already have?

Mechanism - why the network matters. Firms mostly hire locally, and few Japanese workers move between prefectures. Skills are partly tied to an industry, but they can be used in industries with similar jobs and tasks. If many local workers already work in industries that are skill-linked to industry *i*, a new firm in *i* can find and train staff more cheaply, so *i* is more likely to appear in that prefecture.

Hypothesis: The more skill links an industry has to industries a prefecture already specialises in, the more likely the prefecture is to become specialised in that industry between 2016 and 2021.

3. Data and method

Nodes and links: Nodes are about 90 industries (JSIC major groups). Two industries are linked if they employ similar types of workers measured with the 2015 Population Census counts of workers by industry and occupation, available on e-Stat.

Outcome: The Economic Census 2016 and 2021 shows which industries each prefecture is specialised in, meaning the industry's share of local jobs is higher than in Japan as a whole. A prefecture enters an industry if it was not specialised in it in 2016 but was in 2021.

Network measure: relatedness density, the share of an industry's links that lead to industries the prefecture is already specialised in (0 = none, 1 = all).

Method: we compare how often prefectures enter industries with high and low density, and confirm this with a logit regression of entry on density, with prefecture and industry fixed effects, for all 47 prefectures × 90 industries; as in Neffke et al. (2017), we also compare the skill network against a co-location network.

Accessibility: All data are free official statistics on e-Stat.

Main risk: similar occupations are only a proxy for shared skills; we check linked industries against job changers in the 2022 Employment Status Survey (a full industry-to-industry move matrix requires microdata on

application). Plan B: real job moves between the survey's 20 broad industries, or the German skill network published by Neffke et al. (2017).

4. Expected results and impact

If the hypothesis holds, prefectures will enter high-density industries clearly more often than low-density ones. If it does not hold, that is also useful: where workers rarely switch jobs, skills may not be the main limit, and policy should look at other factors such as supplier links.

Impact - for whom and how. (1) *Prefectural governments*, especially where manufacturing or retail jobs are disappearing, get a ranked list of realistic next industries based on skills local workers already have, so less public money goes to industries that cannot find staff. (2) *Public job centres and training programmes* can retrain people for industries one step away in the skill network, making training shorter and re-employment more likely. (3) *Workers* in shrinking industries see where their skills are useful.

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